PES University — Dept. of CSE

SOFTWARE ENGINEERING LAB 1

Requirements Engineering & UML Use-Case Modelling

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Problem Statement #63 — Sustainability & Green Tech

Community Solar Credit Allocation Manager

A clean energy sharing portal that ingests rooftop solar generation metrics, facilitates peer-to-peer solar credit transfers within a neighborhood microgrid, and tracks monthly utility billing offsets.

Stakeholders / Actors: Prosumer Resident, Co-op Manager, Smart Meter Gateway, Co-op Billing Ledger

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Requirements Table

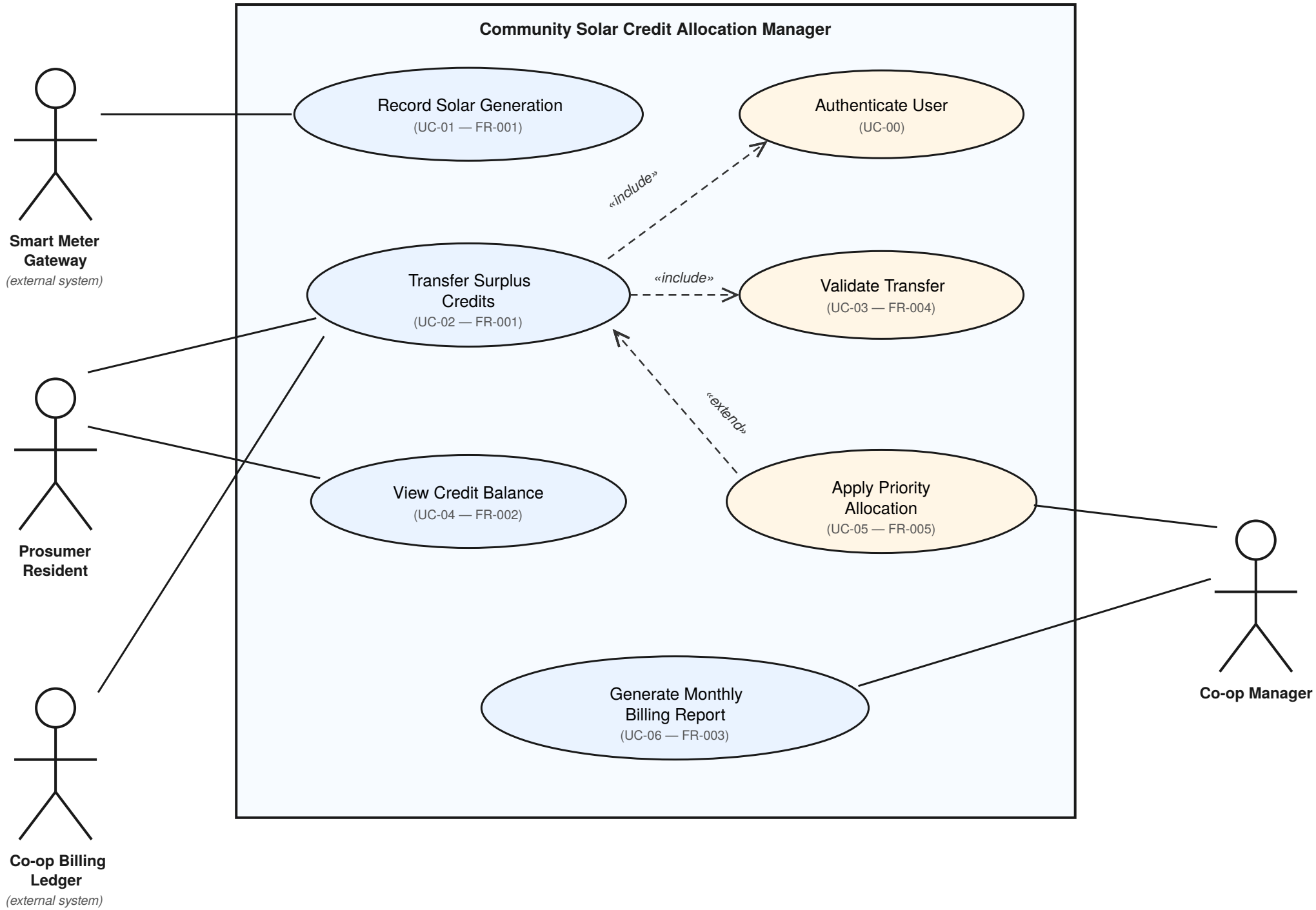
Community Solar Credit Allocation Manager — Problem Statement #63 (Sustainability & Green Tech)

Stakeholders / Actors: Prosumer Resident, Co-op Manager, Smart Meter Gateway, Co-op Billing Ledger (external systems)

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall record solar generation units (kWh) from smart meter feeds and allow prosumers to transfer surplus credits to neighbor accounts.	High	Pass: Transferred credits deduct from the prosumer's balance and apply to the neighbor's billing statement. Fail: A transfer that exceeds the prosumer's available balance is rejected.	Enables the core peer-to-peer credit-sharing purpose of the microgrid portal.
FR-002	Functional	The system shall allow a Prosumer Resident to view their real-time solar credit balance and a chronological log of their generation and transfer transactions.	Medium	Pass: Opening the “My Balance” screen displays a balance figure and transaction list that matches the backend ledger within 5 seconds. Fail: Balance shown differs from the ledger or fails to load.	Gives residents transparency and trust in how their generated credits are being tracked and used.
FR-003	Functional	The system shall allow a Co-op Manager to generate a monthly billing offset report summarizing the solar credits applied to each household's utility statement.	High	Pass: Running the report for a given month produces one row per household with generation, transfers, and net offset that sums to the utility invoice adjustment. Fail: Totals do not reconcile with individual transactions.	Required for the co-op to reconcile credits against the utility's monthly billing cycle.
FR-004	Functional	The system shall validate every credit-transfer request against the prosumer's smart-meter-verified surplus balance before authorizing the transfer («include» of Transfer Surplus Credits).	High	Pass: A transfer request within the verified surplus balance is authorized within 2 seconds. Fail: A request exceeding the verified balance is blocked with an on-screen error before any ledger update occurs.	Protects ledger integrity by ensuring no household can transfer more credit than it has actually generated.
FR-005	Functional	The system shall optionally apply a priority allocation rule that routes a configurable share of transferred surplus credits to designated low-income or community accounts first («extend» of Transfer Surplus Credits).	Low	Pass: When priority allocation is enabled for a transfer, the configured share of credits is routed to the designated accounts before the remainder reaches the intended recipient. Fail: Priority accounts receive no credit despite the rule being enabled.	Supports the co-op's equity goals by letting managers favor vulnerable households during surplus periods, without complicating the standard transfer flow.
NFR-001	Nonfunctional (Performance & Security)	The smart meter telemetry processing pipeline must validate and store hourly generation data for 500 households with 99.9% uptime.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load. Fail: Uptime drops below 99.9% or ingestion latency exceeds the agreed threshold during load testing.	A reliable, always-available ingestion pipeline is essential since billing offsets depend on accurate hourly readings.
NFR-002	Nonfunctional (Security & Auditability)	The system shall encrypt all smart-meter telemetry and credit-transfer transactions in transit and at rest (TLS 1.2+/AES-256), and shall record every transfer in a tamper-evident audit log retained for at least 7 years.	High	Pass: A security scan confirms encryption in transit and at rest, and every test transfer produces an immutable, timestamped audit entry. Fail: Any transaction is found unencrypted or missing from the audit log.	Protects sensitive household energy-usage data and gives the co-op and utility a defensible record for billing disputes.

Note: FR-001 and NFR-001 are the instructor-provided sample requirements. FR-002 through FR-005 and NFR-002 were authored for this problem statement.

Community Solar Credit Allocation Manager — Use-Case Diagram



Use-Case Flow Specification

Community Solar Credit Allocation Manager — Problem Statement #63

Use Case: Transfer Surplus Credits (UC-02)

Primary Actor: Prosumer Resident

Supporting Actors / Systems: Smart Meter Gateway (provides verified generation/balance data); Co-op Billing Ledger (receives applied credits and reflects them on the neighbor's utility statement)

Related Requirements: FR-001, FR-004 («include» Validate Transfer), FR-005 («extend» Apply Priority Allocation), «include» Authenticate User

Preconditions

- The Prosumer Resident is a registered account holder and has successfully authenticated («include» Authenticate User).
- The prosumer's rooftop generation data has been recorded by the Smart Meter Gateway, and a positive surplus credit balance exists.
- The recipient neighbor account exists and is active within the same microgrid co-op.

Postconditions

- Success: The transferred credit amount is deducted from the prosumer's balance and applied to the neighbor's billing statement; the transaction is written to the audit log and both parties are notified.
- Failure: No balances are changed and no transaction is recorded; the prosumer is shown an explanatory error.

Main Success Scenario

1. Prosumer Resident logs into the portal and selects “Transfer Surplus Credits” («include» Authenticate User).
2. System displays the prosumer's current surplus balance and a transfer form (neighbor account ID, credit amount).
3. Prosumer Resident enters the neighbor's account ID and the desired credit amount, then submits the request.
4. System validates the request against the smart-meter-verified surplus balance («include» Validate Transfer).
5. System checks whether a priority allocation rule is active for the transfer («extend» Apply Priority Allocation); if active, the configured share is routed to designated accounts first.
6. System deducts the transferred credits from the prosumer's balance.
7. System applies the (remaining) credits to the neighbor's billing statement via the Co-op Billing Ledger.
8. System logs the transaction in the audit trail and sends a confirmation notification to both the prosumer and the neighbor.
9. Use case ends successfully.

Alternate Flow 4a: Insufficient Balance

a) At step 4, System detects that the requested transfer amount exceeds the prosumer's verified surplus balance.
b) System displays an error message: “Transfer exceeds available balance.”
c) System prompts the Prosumer Resident to either reduce the transfer amount or cancel the request.
d) If the prosumer cancels, the use case ends without any change to balances or the billing statement; if the prosumer revises the amount, the flow resumes at step 3.